

LOCAL LANGLANDS CORRESPONDENCE FOR $\mathrm{GL}_n/\mathbb{R}$

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ABSTRACT. These are the notes for my short talk for the finals of Geometric Representation Theory (M392C) taught by Lucas Mason Brown.

1. INTRODUCTION

Let F be a local field. Roughly, the Local Langlands Correspondence (LLC) for GL_n/F is a bijection between the n -dimensional (+ adjectives) representations of the Weil group W_F of F [Galois side] and the irreducible (+ adjectives) representations of $\mathrm{GL}_n(F)$ [Automorphic side]. The Weil group of a local field is a modification of its absolute Galois group. In this note, we will describe the LLC for the local field $\mathbb{R}$. In fact, the archimedean case motivated Langlands to consider such potential relations over non-archimedean local fields, leading to the formulation of the LLCs!

The representation theory of $W_{\mathbb{R}}$ is quite simple. On the other hand, the classification of irreducible admissible representations of a real reductive group upto infinitesimal equivalence was achieved by Langlands ([3]), and is called the Langlands classification. The LLC for $\mathrm{GL}_n/\mathbb{R}$ is evident from the Langlands classification for GL_n , once the representations of $W_{\mathbb{R}}$ are identified.

2. THE WEIL GROUP OF $\mathbb{R}$

For any local field F , one can define the Weil group W_F axiomatically (see [4]). The Weil group for $\mathbb{R}$ is a non-split extension of $\mathrm{Gal}(\mathbb{C}/\mathbb{R})$ by $\mathbb{C}^\times$. Concretely, we have

$$W_{\mathbb{R}} = \mathbb{C}^\times \cup j\mathbb{C}^\times,$$

where $j^2 = -1$, $jzj^{-1} = \bar{z}$ for $z \in \mathbb{C}^\times$. These are the units of the quaternion algebra: the non-split extension is given by the non-trivial element of the Brauer group of $\mathbb{R}$ (which is $\mathbb{Z}/2\mathbb{Z}$), and this element corresponds to a non-split central simple algebra over $\mathbb{R}$ - the quaternions. We will however, work with the concrete description.

3. REPRESENTATIONS OF $W_{\mathbb{R}}$

In this section, we will describe the representations of $W_{\mathbb{R}}$. We will only consider (complex) representations of $W_{\mathbb{R}}$ where the group elements act semisimply.

3.1. One dimensional representations. The characters of $W_{\mathbb{R}}$ factor through its abelianization. One computes immediately the following:

$$jzj^{-1}z^{-1} = \bar{z}z^{-1},$$

so the commutator subgroup is the unit circle S^1 in $\mathbb{C}^\times$. $\mathbb{C}^\times/S^1$ is the positive real numbers, and we can identify $j\mathbb{C}^\times/S^1$ with the negative reals. Therefore, $W_{\mathbb{R}}^{\mathrm{ab}} = \mathbb{R}^\times$. This is one of the axioms in [4].

The (continuous) characters of $\mathbb{R}^\times$ are just $x \mapsto x^s$ for $s \in \mathbb{C}$. The continuous characters of $W_{\mathbb{R}}^\times$ are thus parameterized by a complex variable s and a character ϵ of $\{\pm 1\}$.

Because of the identification $W_{\mathbb{R}}^{\mathrm{ab}} \simeq \mathbb{R}^\times$, we have

Theorem 3.1. *[LLC for $\mathrm{GL}_1/\mathbb{R}$] There is a bijection between irreducible (admissible) representations of $\mathrm{GL}_1(\mathbb{R})$ and the one dimensional (semisimple) representations of the Weil group $W_{\mathbb{R}}$.*

In general, any character of $\mathbb{C}^\times$ is given by a pair of complex numbers μ, ν ,

$$z \mapsto z^\mu \bar{z}^\nu,$$

where the only restriction is that $\mu - \nu$ is an integer. We denote this character by $\chi_{\mu, \nu}$.

3.2. Two dimensional representations. Here, we identify the two dimensional irreducible representations of $W_{\mathbb{R}}$. As before, we assume that the elements of $W_{\mathbb{R}}$ act semisimply.

Let φ be a two dimensional representation of $W_{\mathbb{R}}$. First, we restrict the representation φ, V to $\mathbb{C}^\times$. The set of operators $\varphi(z), z \in \mathbb{C}^\times$ is a family of commuting semisimple operators, so we choose a basis u, v of V in which the matrix representation is diagonal and write

$$\varphi = \begin{pmatrix} \chi_{\mu, \nu} & 0 \\ 0 & \chi_{\mu', \nu'} \end{pmatrix}.$$

Let $u' = \varphi(j)u$. For $z \in \mathbb{C}^\times$, we compute

$$\varphi(z)(u') = \varphi(j\bar{z}j^{-1})(u') = \chi_{\mu, \nu}(\bar{z})\varphi(j)(u) = z^\nu \bar{z}^\mu u'.$$

So u' is also a common eigenvector for the $\mathbb{C}^\times$ action. If the two characters $\chi_{\mu, \nu}$ and $\chi_{\mu', \nu'}$ are the same, then the representation is reducible - and decomposes as sum of eigenspaces of the semisimple operator $\varphi(j)$. If u' is a scalar multiple of u , then $\mu = \nu$, and similar reasoning for $v' = \varphi(j)v$ shows that $v' \in \mathbb{C}v$ and $\mu' = \nu'$. Again, this representation decomposes as a sum of two characters of $W_{\mathbb{R}}$.

Finally suppose that $u' \in \mathbb{C}v$ and that φ is irreducible. In this case, we have $\mu = \nu'$ and $\mu' = \nu$ (and $\mu \neq \nu$). The representations restricting to $\chi_{\mu, \nu} \oplus \chi_{\nu, \mu}$ and $\chi_{\nu, \mu} \oplus \chi_{\mu, \nu}$ are isomorphic. Therefore, irreducible two dimensional representations of $W_{\mathbb{R}}$ are parameterized by a complex number $t = \frac{1}{2}(\mu + \nu)$ and a positive integer $l = \mu - \nu$. We denote the associated representation by $\pi_{l, t}$.

3.3. Reducibility of representations. Let φ, V be a n -dimensional representation of $W_{\mathbb{R}}$. Consider the representation V as a $\mathbb{C}^\times$ representation, and one has the decomposition into (one dimensional) eigenspaces

$$V = \oplus_{\mu, \nu} V_{\mu, \nu}.$$

As before, one sees that the action of $\varphi(j)$ permutes the eigenspaces. If $\varphi(j)V_{\mu, \nu} = V_{\mu, \nu}$ then it is a one dimensional subrepresentation of V . Otherwise, $\varphi(j)V_{\mu, \nu} = V_{\nu, \mu}$, and as before, the representation $V_{\mu, \nu} \oplus V_{\nu, \mu}$ is representation of $W_{\mathbb{R}}$.

Theorem 3.2. *The n dimensional representations of $W_{\mathbb{R}}$ are completely reducible, and each irreducible constituent is either one or two dimensional.*

4. THE LANGLANDS CLASSIFICATION FOR $\mathrm{GL}_n(\mathbb{R})$

In this section, we state Langlands classification for irreducible admissible representations of $\mathrm{GL}_n(\mathbb{R})$ upto infinitesimal equivalence. As mentioned earlier, the representations of $\mathrm{GL}_1(\mathbb{R})$ are parameterized by a complex variable s and a character of $\{\pm 1\}$. We will first mention some special representations of $\mathrm{GL}_2(\mathbb{R})$ and then talk about how one builds representations of GL_n from these representations in rank 1 and 2.

4.1. Special Representations of GL_2 . For $l \geq 2$ an integer, recall that $\mathrm{SL}_2(\mathbb{R})$ has the discrete series representations D_l^+ , and the limit of discrete series representations denoted D_1^+ . Let $\mathrm{SL}_2^\pm(\mathbb{R})$ denote the group of 2×2 real matrices with determinant ± 1 . We consider the following (tempered) representations of $\mathrm{SL}_2^\pm(\mathbb{R})$:

$$D_l = \mathrm{Ind}_{\mathrm{SL}_2(\mathbb{R})}^{\mathrm{SL}_2^\pm(\mathbb{R})}(D_l^+),$$

for $l \geq 1$.

Next, consider the *relative discrete series* representations $\sigma_{l, t}$ of $\mathrm{GL}_2(\mathbb{R})$, given by

$$\sigma_{l, t} = D_l \otimes |\det|^t,$$

where $l \geq 1$ and $t \in \mathbb{C}$. These representations will be the building blocks (in rank 2) for representations of $\mathrm{GL}_n(\mathbb{R})$. (Relate this to the representation

4.2. Langlands Classification for GL_n . We state Langlands' classification of irreducible admissible representations of $\mathrm{GL}_n(\mathbb{R})$ upto infinitesimal equivalence. There is a more general version for an arbitrary reductive group in place of GL_n , but we will not state that here (see [1], Chapter 4).

For any partition of $n = \sum_{i=1}^r n_i$ where $n_i = 1$ or 2 , and representations σ_i of GL_{n_i} of the form mentioned above (i.e. characters of $\mathbb{R}^\times$ or the relative discrete series $\sigma_{l,t}$), let M be the Levi subgroup $\prod_i \mathrm{GL}_{n_i}(\mathbb{R})$ of $\mathrm{GL}_n(\mathbb{R})$ and let P be the standard parabolic of $\mathrm{GL}_n(\mathbb{R})$ containing M . Inflate the representation $\sigma = (\sigma_1, \dots, \sigma_r)$ of M to P , and consider the normalized induction

$$I(\sigma_1, \dots, \sigma_r) = \mathrm{Ind}_P^{\mathrm{GL}_n(\mathbb{R})} ((\sigma_1, \dots, \sigma_r) \otimes \rho_P),$$

where ρ_P denotes the half-sum of positive roots associated to the parabolic P .

Theorem 4.1 (Langlands). (1) Suppose $t_1/n_1 \geq t_2/n_2 \geq \dots \geq t_r/n_r$, then $I(\sigma_1, \dots, \sigma_r)$ has a unique irreducible quotient denoted $J(\sigma_1, \dots, \sigma_r)$.
(2) All irreducible admissible representations of $\mathrm{GL}_n(\mathbb{R})$ are infinitesimally equivalent to some $J(\sigma_1, \dots, \sigma_r)$.
(3) $J(\sigma_1, \dots, \sigma_r)$ and $J(\sigma'_1, \dots, \sigma'_{r'})$ are infinitesimally equivalent if and only if $r = r'$ and the σ'_i are a permutation of σ_i (such that the condition in (1) holds).

Corollary 4.2 (LLC for $\mathrm{GL}_n/\mathbb{R}$). There is a one-one correspondence between irreducible admissible representations of $\mathrm{GL}_n(\mathbb{R})$ upto infinitesimal equivalence and the isomorphism classes of n -dimensional representations of $W_{\mathbb{R}}$ where the group acts semisimply.

Proof. Let $J = J(\sigma_1, \dots, \sigma_r)$ be an irreducible admissible representation of $\mathrm{GL}_n(\mathbb{R})$. To each representation σ_i of GL_{n_i} , consider the associated n_i -dimensional Weil group representation: if $n_i = 1$, then consider the character $\tau_i : W_{\mathbb{R}} \rightarrow \mathbb{R}^\times \rightarrow \mathbb{C}^\times$, where the first map is the projection $W_{\mathbb{R}} \rightarrow W_{\mathbb{R}}^{\mathrm{ab}}$ and the second map is σ_i (τ_i corresponds to σ_i in Theorem 3.1). If $n_i = 2$, consider the representation $\tau_i = \tau_{l,t}$ described in Section 3, where l, t are the parameters associated to σ_i . Now associate to J the n -dimensional representation $\oplus_i \tau_i$ of $W_{\mathbb{R}}$.

This association is the desired bijection. The inverse map is clear: to any n -dimensional Weil representation V , consider its decomposition $V = \sum_{i=1}^r \tau_i$ into one and two dimensional pieces. For each i , consider the representation σ_i of $\mathrm{GL}_1(\mathbb{R})$ or $\mathrm{GL}_2(\mathbb{R})$ with the same parameters as τ_i as above. Permute the factors so that the condition (1) of the theorem is satisfied, and consider $J(\sigma_1, \dots, \sigma_r)$. $\square$

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